

UNIT 4



DATA COMMUNICATION AND COMPUTER NETWORK



Q.1: LIST THE PROPERTIES OF A GOOD COMMUNICATION SYSTEM. EXPLAIN ANYONE.

PROPERTIES OF A GOOD COMMUNICATION SYSTEM:

The effectiveness of a data communications system depends on the fundamental characteristics which include delivery, accuracy and timeliness.

(i) Delivery:

Making sure that the data is delivered is the first fundamental characteristic of any communication network. The system must be able to deliver data in correct order to the correct destination.

(ii) Accuracy:

System must deliver the data accurately. Data that has been altered during transmission and left uncorrected is not useful.

(iii) Timeliness:

The data must be delivered in a timely manner. Late delivered data is useless.

Q.2: DEFINE TOPOLOGY. WHAT DOES THE TOPOLOGY OF NETWORK DESCRIBE?

TOPOLOGY:

The physical layout in which computers are connected is called topology. The topology is called network describes the way computers are connected. Topology is a major design consideration for computer networking.

Q.3: DESCRIBE MERITS AND DEMERITS OF DIFFERENT TOPOLOGIES OF NETWORK.

BUS TOPOLOGY:

As name suggests, in Bus Topology computers and other devices are connected with a single cable. The central cable is the backbone of the network and every device communicates with the other device through this bus. The advantages of Bus Topology are simplicity, low cost and easy expansion of the network. The disadvantage of the Bus Topology is that a breakdown in the bus cable brings the entire network down.

RING TOPOLOGY:

In Ring Topology, computers are connected in a ring or circle shape. The signal travels around the loop in one direction and passes through each computer. The recipient of the message receives the message while another computer acts like a repeater to send it to the next computer. The failure of a link or a computer can make the entire network non-functional.

STAR TOPOLOGY:

In a star topology all the computers are connected to a central device called hub or switch. To communicate with any computer, the sender must send information to the hub. Then the hub transmits that information to the destination. The advantages of star topology are easy to set up and easy expansion of the network. Another feature of Star topology is that if one link to the hub breaks, only the station using that link is affected not the whole network. It requires more cable length than a bus topology. If the connecting network device (network switch) fails, nodes attached are disabled and cannot participate in computer network communication.

Q.4: DIFFERENTIATE BETWEEN ANALOG AND DIGITAL SIGNALS.

ANALOG SIGNALS	DIGITAL SIGNALS
AN ANALOG SIGNAL IS A CONTINUOUS WAVE THAT CHANGES BY TIME TO PERIOD.	DIGITAL SIGNAL IS A DISCRETE WAVE THAT CARRIES INFORMATION IN A BINARY TERM.
ANALOG SIGNAL HAS NO NEEDS FIXED RANGE.	DIGITAL SIGNAL HAS A FINITE NUMBER I.E. 0 AND 1.
AN ANALOG SIGNALS CAN EASILY BE DISTURBED BY OTHER SIGNAL OR WAVE.	DIGITAL SIGNALS IS LESS PRONE BY OTHER SIGNALS DISTURBANCE.
THE HUMAN VOICE IS AN EXAMPLE OF AN ANALOG SIGNAL.	SIGNAL USED BY COMPUTER ARE DIGITAL SIGNALS.
AN ANALOG SIGNALS ARE REPRESENTED BY A SINE WAVE.	DIGITAL SIGNALS ARE REPRESENTED BY SQUARE WAVES.

Q.5: DIFFERENTIATE BETWEEN DATA RATE OR BIT RATE?

DATA RATE	BAUD RATE
BIT RATE TELLS THE NUMBER OF BITS TRANSMITTED PER UNIT TIME (SECONDS).	BAUD RATE IS USED WHEN WE WANT TO KNOW THE NUMBER OF SIGNAL UNIT TRANSMITTED PER UNIT OF TIME. (SECONDS)
BIT RATE IS THE NUMBER OF BITS (I.E 0S AND 1S) TRANSMITTED PER UNIT TIME.	BAUD RATE IS THE NUMBER OF TIMES A SIGNAL IS TRAVELING COMPRISED OF BITS. ONE SIGNAL CAN REPRESENT MORE THAN ONE BIT.
$\text{BIT RATE} = \text{BAUD RATE} \times \text{THE NUMBER OF BITS PER SIGNAL UNIT.}$	$\text{BAUD RATE} = \text{BIT RATE} / \text{THE NUMBER OF BITS PER SIGNAL UNIT}$

Q.6: EXPLAIN COMPONENTS OF COMMUNICATION SYSTEM USING SIMPLE EXAMPLE.

COMPONENTS OF A COMMUNICATION SYSTEM:

A Communication system has following five components as shown in the given figure.

(i) Message:

It is the information or data to be communicated. Common forms of information include text, numbers, pictures, audio and video.

(ii) Sender:

It is the device that generates and sends a message. It can be a computer, telephone handset, etc.

(iii) Receiver:

It is any particular digital electronic device which has capability to receive data in form of message. The location of receiving computer is generally different from the sending computer. Like sender, it can also be a computer, telephone handset, etc.

(iv) Medium

It is the channel or path through which the message is carried from sender to the receiver. Some examples include twisted-pair cable, coaxial cable, radio waves, etc.

(v) Protocol

Protocols are the rules and procedures on which computers exchange data on network. Sender and receiver follow same protocols to communicate with each other. In other words, a protocol is an agreement between two parties or venders, using communication devices.

Q.7: WHAT IS NETWORK ARCHITECTURE?

NETWORK ARCHITECTURE:

It is the design of a computer network. It is a framework for the specification of a network's physical components, their functional organization and configuration, operational procedures and communication protocols used. Just like OSI / TCP layered architecture.

Q.8: WHAT IS A NETWORK ADDRESS?

NETWORK ADDRESS:

Network addresses are like our house addresses. They must be unique and distinctive. This avoids confusion for the postman. A network address is any Logical or Physical Address that uniquely identifies it from others. This address is needed to distinguish a network node or device on a computer network. It is a numeric or symbolic number or address that is assigned to any device that seeks access to network or is the part of a network. We have to remember, Physical and Logical Address are different.

Q.9: DESCRIBE GUIDED MEDIA AND ITS TYPES.

GUIDED MEDIA:

If In guided media signals are transmitted in a narrow pathway by using physical links. It is also called Wired or Bounded transmission media. The physical links are the cables that are tangible or have physical existence. There are three common types of guided media used for the networks. Each of them has its own characteristics like transmission speed, effect of noise, physical appearance, cost, etc.